Convex optimization for machine learning – TD3

Olivier Fercoq
Bruno Costacèque

January 2022

1 Exercises

Exercise 1. (Proximal gradient for logistic regression)

We consider a classification problem defined by observations $(\mathbf{x}_i, y_i)_{1\leq i\leq n}$ where for all i, $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{-1, 1\}$. The coordinates of $\mathbf{x}_i$ are the explanatory variables of the model. We note $x_{i,j}$ the i-th observation of the j-th explanatory variable, so that $\mathbf{x}_i = (x_{i,1}, \cdots, x_{i,p})$.

We propose the following linear model for the generation of the data. Each observation is supposed to be independent and there exists a vector $\mathbf{w} \in \mathbb{R}^p$ and $w_0 \in \mathbb{R}$ such that for all i, $(y_i, \mathbf{x}_i)$ is a realization of the random variable $(Y, \mathbf{X})$ whose law satisfies

$$\mathbb{P}_{w_0,\mathbf{w}}(Y = 1|\mathbf{X}) = \frac{\exp(\mathbf{X}^\top \mathbf{w} + w_0)}{1 + \exp(\mathbf{X}^\top \mathbf{w} + w_0)} \,.$$

1. In this model, what are the explanatory variables? The observed outcomes? The parameters? Can you think of a real-life situation to which you could apply this model?

2. We set $\sigma := x \mapsto (1 + \exp(-x))^{-1}$; this function is known as the *logistic function*.

   Show that $\forall i \in [\![1, n]\!]$, $\mathbb{P}_{w_0,\mathbf{w}}(Y_i = y_i|\mathbf{x}_i) = \dfrac{1}{1 + \exp(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0))} = \sigma\big(y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)$.

3. Show that the maximum likelihood estimator is

$$(\widehat{w_0}, \widehat{\mathbf{w}}) = \arg\min_{w_0,\mathbf{w}} \sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big)$$

4. Denote $f(w_0, \mathbf{w}) = \sum_{i=1}^n \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big)$. Compute $\nabla f(w_0, \mathbf{w})$.

5. Compute the proximal operator of $\mathbf{x} \mapsto \frac{\lambda}{2}\|\mathbf{x}\|^2$.

6. Write the proximal gradient method for the logistic regression problem with ridge regularizer

$$(\widehat{w_0}^\lambda, \widehat{\mathbf{w}}^\lambda) = \arg\min_{w_0,\mathbf{w}} \sum_{i=1}^{n} \log\Big(1 + \exp\big(-y_i(\mathbf{x}_i^\top \mathbf{w} + w_0)\big)\Big) + \frac{\lambda}{2}\|\mathbf{w}\|^2.$$

7. Compute the Hessian matrix and write Newton's method for the same problem.

Exercise 2. (LASSO)

We consider the problem

$$\min_{\mathbf{x}\in\mathbb{R}^n} \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda\|\mathbf{x}\|_1 \,.$$

1. Prove that the solution is $\{0\}$ for large λ.

2. For an arbitrary λ, provide the expression of the proximal gradient algorithm, using the step size $\gamma_k := 1/L$.

3. Assume that the initial point is at distance D from a minimizer. How many iterations are needed (at most) to achieve an ε-minimizer?